What is a Pod in Kubernetes?

A **Pod** is the smallest and simplest unit in the Kubernetes object model. It represents a **single instance of a running process** in your cluster.

Each pod contains:

- One or more **containers** (usually one)
- Shared **storage/network**
- Information about how to run the containers (e.g., image, ports, environment variables)

Think of a pod as a wrapper around containers to manage and run them together as a single unit.

Why use a Pod?

- Multiple tightly-coupled containers can share the same resources and run together.
- All containers in a pod share the **same IP address**, **hostname**, and **storage volumes**.

Example: Simple Pod YAML with a single container

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
spec:
  containers:
  - name: nginx-container
    image: nginx:latest
    ports:
    - containerPort: 80
```

Explanation:

- **apiVersion**: The version of the Kubernetes API.
- **kind**: The type of object (here, Pod).
- **metadata.name**: The name of the pod.
- **spec.containers**: A list of containers in the pod.
    - **name**: The name of the container.
    - **image**: The Docker image used.
    - **containerPort**: The port the container listens on.

How to Create the Pod

Save the above YAML as nginx-pod.yaml and run:

kubectl apply -f nginx-pod.yaml

Then check the pod status:

kubectl get pods

To describe the pod:

Kubectl describe pod <podname>

To delete the pod:

kubectl delete -f nginx-pod.yaml